Journal of Physics:
Conference Series

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To cite this article: Filip Gषृrski *et al* 2022 *J. Phys.: Conf. Ser.* **2198** 012040

Journal of Physics: Conference Series **2198** (2022) 012040 doi:10.1088/1742-6596/2198/1/012040

Automated Design and Rapid Manufacturing of Low-Cost Customized Upper Limb Prostheses

Filip Górski ,Przemysław Zawadzki, Radosław Wichniarek, Wiesław Kuczko, Sandra Słupińska, Magdalena Żukowska

Faculty of Mechanical Engineering, Poznan University of Technology, Piotrowo 3 STR, 61-138 Poznan, Poland

Corresponding author. Tel.: +48 61 665 2708; fax: +48 61 665 2774. E-mail address: filip.gorski@put.poznan.pl

Abstract. The paper presents results of studies on an innovative approach to mass customization of prostheses, namely automated design and rapid manufacturing (3D printing), on the basis of 3D scanning of upper limb of a given patient. Two different products are presented – a simple mechanical prosthesis and a cosmetic prosthesis. In both cases, intelligent CAD models , with adaptive geometry – changing automatically to the data from 3D scanning – were prepared. The digital models were built and tested using data of real patients. A special algorithm of 3D scanning data processing was prepared and implemented, as well as a set of algorithms of automated data processing for 3D printing processes. The test models were manufactured using low-cost 3D printing technology. The verification was successful – possibility of preparing customized prostheses was reached, using automated design.

1. Introduction

Lack of one or more upper limbs is a very serious disability, resulting in reduced possibility of performing most everyday activities, as well as possible number of other health problems. It is a problem that concerns a quite significant amount of people every year. The missing upper limb or its part can be a result of a birth defect or an amputation. The amputations are most frequently performed because of vascular diseases, diabetes and injuries.

A missing limb is not only a hindrance in day-to-day life, but also makes for a significant psychic discomfort. That is why it is very important to allow patients to have a possibility to obtain a well-working replacement. Biomedical engineers continuously work on development and improvement of designs of prostheses, to make them naturally replace the missing limb and not cause additional discomfort. Appropriately designed geometry of the prosthesis should maximize extent of movement, ensure balance of weight and stability of the prosthesis, as well as guarantee comfort of prolonged use [1].

Present level of technological advancement allows manufacturing prostheses of upper limbs, which can replace the lacking limb, to a certain degree. Their functions may be purely visual (cosmetic prostheses), but they can also be fully operational – controlled mechanically or electronically. A prosthesis, as an artificial replacement of a missing body part, is usually manufactured in several steps, involving manual shaping of prosthetic socket on the basis of measurements of patient's stump. As such, the production should be considered as single piece production. The total manufacturing time can take, depending on the particular prosthesis type, anywhere between one week and several months [2].

For a typical patient, a problem in accessing these devices is their price, which is proportional to technological advancement and quality of production of a given prosthesis. Time of obtaining a prosthesis is also an important factor, especially in case of injuries and small children, where several weeks or even months can be much too long both from therapeutic and psychic comfort of the patient point of view. It is also important that fitting prosthesis to a given patient is not a simple task. As certain studies indicate, there is often a problem in mutual communication between a patient and a prosthetist and it can negatively affect final satisfaction of use of a given prosthesis [3].

Medical products, such as orthopaedic supplies, implants and prostheses are more and more often manufactured using additive manufacturing (3D printing) technologies [4-5]. The technology of layered manufacturing (another name for 3D printing) is invaluable when customized shapes must be produced, as no tooling is required. The most popular process of Fused Deposition Modelling (FDM, also known as Fused Filament Fabrication, FFF) is very cheap both in terms of machines and thermoplastic materials available. Therefore, it is becoming increasingly popular in various medical applications [6,7].

One of the largest problems in 3D printing of customized orthopaedic supplies is requirement of specialized engineering knowledge. The patient anthropometric data must be gathered and processed, usually manually. This can generate a lot of inaccuracies [8]. Obtaining a shape requires many hours of advanced surface modelling in CAD systems. Additionally, 3D printing of thermoplastic products with satisfying values of accuracy and strength is difficult, as process parameters significantly influence properties of obtained parts [9]. That is why a traditional process of making plaster casts has still not been replaced with 3D printing. There are constant studies on how to make the data gathering, processing and manufacturing easier and more available in general medical practice. Automation of certain engineering tasks seems a promising direction [10].

The recent, ongoing transition from a traditional, manual process of manufacturing prostheses into making them by 3D printing, semi-automatically on the basis of anatomical measurements, allows to conclude that production paradigm has changed from piece production into mass customization. Mass customization is making individualized, customizable products on a mass scale. The concept is known since the 70s of the XXth century and was mostly introduced in automotive and aeronautical industries [11,12]. Mass customization (MC) often relies on Knowledge Based Engineering (KBE) systems, intelligent CAD models and elastic production systems [13,14].

In orthopaedics, MC would seem as a required, necessary approach, as each patient is different and the demand is potentially massive. Only a minor part of patients actually use upper limb prostheses on a daily basis. The authors of this paper performed a series of interviews with both patients and doctors to investigate on this topic, as a part of preliminary studies. The main problem seems to be long time of delivery (several weeks, even up to several months) and high cost of prostheses (often equal to a cost of a new car), which makes the patient give up on the prosthesis and try to live without it. This is especially true in the case of child patients, who would require frequent replacements due to growth. That is why it would be required to shorten time of delivery and decrease costs, while maintaining usefulness of prostheses. The authors aim at this in their R&D work and this paper shows some advancement – namely, two cases of upper limb prostheses designed automatically and produced by means of additive manufacturing (3D printing). The aim of the work was to build such models and prove that it is feasible to build useful prostheses on their basis.

2. Materials and methods

2.1. Aim of the work

The undertaken research and practical work described in this paper were focused on introduction of a new approach in design and production of low-cost upper limb prostheses. The new approach consists in 3D scanning of patient's stump and the other healthy limb (if present), feeding the data into an intelligent CAD model and 3D printing the obtained result. In general, this should be considered as transforming the typical manual single piece production of prostheses into mass customization, allowing much more faster response time and possibility of delivering finished products in mere days, not weeks or months as in the traditional approach.

The paper focuses on two examples of prostheses: the cosmetic prosthesis with a new design of a prosthetic socket and the mechanical prosthesis, based on a popular open source designs – RoboHand and UnLimbited Arm. The two cases were both tested on data of various patients, the design and manufacturing process was performed for two different patients. The main aim was to test whether this approach is feasible, both in terms of obtaining an anatomically correct prosthesis and doing so quickly and cheaply.

The methodology was based on the scheme presented in Fig. 1. The patients were first 3D scanned, then the data was semi-automatically processed. The 3D scanning process methodology was assumed as in the previous works [2,10] – David SLS-3 low-cost optical scanner was used, with a special track and mount built for easier positioning around the patient. The prepared intelligent CAD models were fed with processed data (in form of point coordinates) and automatically adapted their geometry to match anatomy of a given patient. The 3D models were then 3D printed using low-cost FDM technology and the resulting products were verified for correctness and comfort of use.

The design process and tools used are presented in scheme in Figure 2. The intelligent CAD models of the prostheses were built in the Autodesk Inventor CAD system. They were fed with patients' data by use of Excel spreadsheet (separate for each patient), which was filled semi-automatically by algorithms of 3D scan data processing using the MeshLab software.

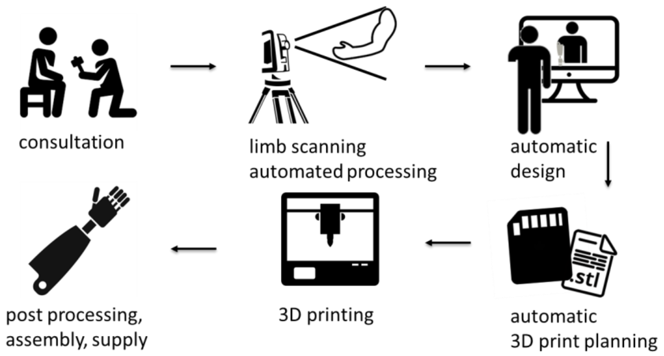


Fig. 1. Methodology of prostheses production – general scheme

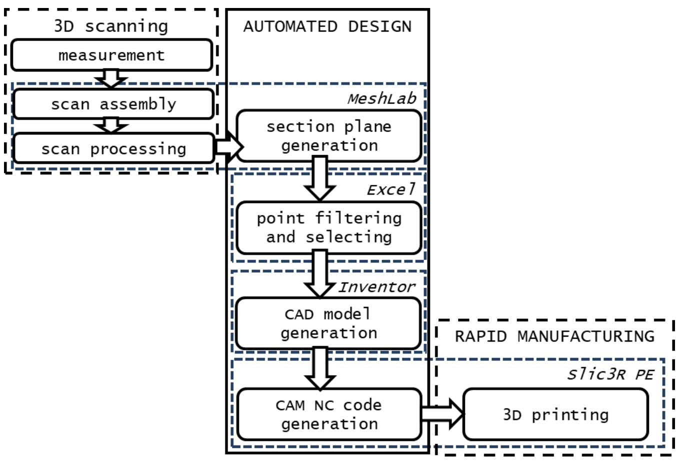


Fig. 2. Methodology of prostheses design and production preparation

2.2. Patient data
The 3D scanning and design process, as mentioned above, were performed for two different patients. It were a 30-year old male patient and an 11-old female patient, both with missing left forearms, albeit various relative lengths of stumps were present (patient 1 – approx. 50% length, patient 2 – approx. 25% length, comparing with a healthy other forearm). The obtained stump geometries (after 3D scanning and semi-automatic processing using MeshLab software with custom macros) are presented in Figure 3.

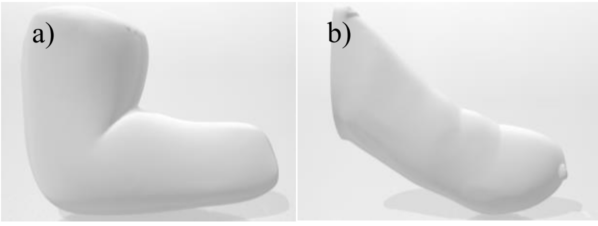


Fig. 3. Data of patients, (a) patient 1 (adult male); (b) patient 2 (child female)

2.3. Cosmetic prosthesis design
As a result of preliminary studies, the cosmetic prosthesis was divided into three parts: prosthetic socket, forearm and hand [2]. Thanks to such a solution, it is possible to adjust functionality of the prosthesis to patient's needs – particular modules can be exchanged. Moreover, thanks to application of a standard threaded mounting, it is possible to use various end effectors – starting from simple prosthetic hands fulfilling esthetic purposes, through mechanical hooks for some manual operations, to (potentially) 3D printed bionic hands, controlled by muscle signals.

However, the most important part of the prosthesis is the prosthetic socket. It decides about patient's comfort and fitting of the prosthesis and is also the most difficult part to automate – in a traditional process it is shaped manually in presence of the patient. The designed geometry is an innovative concept, joining two known designs: TRAC (Transradial Anatomically Contoured) [15] and CRS (Compression/Release Stabilized) [16] prosthetic sockets.

The whole concept is based on the idea of partial compression of patient's soft tissues in some parts of the stump area, simultaneously leaving space for natural release of these tissues in other parts. The TRAC concept specifies anatomical areas that are released (infracubital compartment, distal radial/ulnar, olecranon region contour and epicondylar region contour) and compressed (antecubital channel, area behind infracubital compartment, infra-olecranon region and others). The CRS concept utilizes 4 longitudinal pockets causing compression and holes causing release of soft tissue [17]. The conceptual design is shown in Fig. 4.

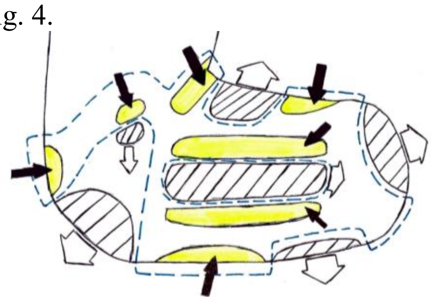
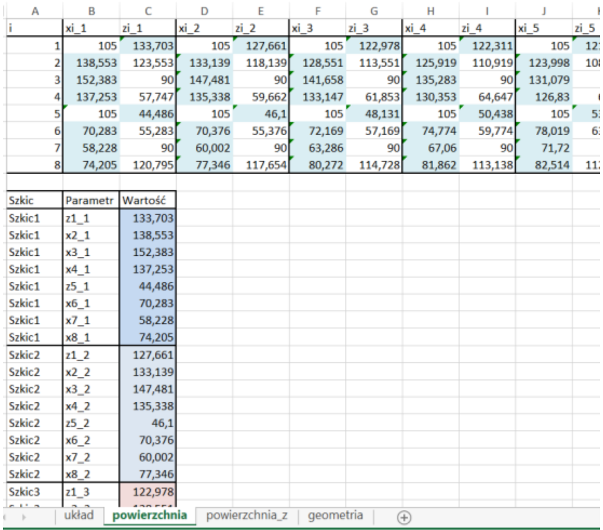


Fig. 4. Own design of prosthetic socket. Black arrows and yellow areas – tissue compression, white arrows and hatched areas – tissue release.

The intelligent CAD model was created in Autodesk Inventor 2017. The data feeding the model is taken from planar sections, performed on a mesh which is a digital representation of 3D scanned stump of a patient. Coordinates of points from the sections are exported to Excel spreadsheet (Fig. 5). There, they are sorted and filtered. Resulting coordinates are sent to a sheet, which is directly connected with the Inventor model (.ipt file) by the iLogic rules, allowing reading values from appropriate cells.



i	xi_1	zi_1	xi_2	zi_2	xi_3	zi_3	xi_4	zi_4	xi_5	zi_5
1	105	133,703	105	127,661	105	122,978	105	122,311	105	[illegible]
2	138,553	123,553	133,139	118,139	128,551	113,551	125,919	110,919	123,998	[illegible]
3	152,383	90	147,481	90	141,658	90	135,283	90	131,079	
4	137,253	57,747	135,338	59,662	133,147	61,853	130,353	64,647	126,83	[illegible]
5	105	44,486	105	46,1	105	48,131	105	50,438	105	[illegible]
6	70,283	55,283	70,376	55,376	72,169	57,169	74,774	59,774	78,019	[illegible]
7	58,228	90	60,002	90	63,286	90	67,06	90	71,72	
8	74,205	120,795	77,346	117,654	80,272	114,728	81,862	113,138	82,514	[illegible]

Szkic	Parametr	Wartość
Szkic1	z1_1	133,703
Szkic1	x2_1	138,553
Szkic1	x3_1	152,383
Szkic1	x4_1	137,253
Szkic1	z5_1	44,486
Szkic1	x6_1	70,283
Szkic1	x7_1	58,228
Szkic1	x8_1	74,205
Szkic2	z1_2	127,661
Szkic2	x2_2	133,139
Szkic2	x3_2	147,481
Szkic2	x4_2	135,338
Szkic2	z5_2	46,1
Szkic2	x6_2	70,376
Szkic2	x7_2	60,002
Szkic2	x8_2	77,346
Szkic3	z1_3	122,978

układ | powierzchnia | powierzchnia_z | geometria

Fig. 5. A spreadsheet containing point coordinates allowing socket generation (partial view)

Seven parallel planes were used to draw closed splines, each based upon 8 points. The number 8 was selected by experiment – preliminary studies allowed to state, that applying a greater number of spline control points (12 and 16), while theoretically more accurate, in practice does not bring an improved result in form of increased accuracy of socket fitting (mostly due to imperfections of the 3D printing process itself).

The coordinates of spline control points are parameterized and taken from the spreadsheet (Fig. 6). Then, the sections are joined through a tool of multi-section extrusion, creating a surface object (Fig. 7a). On the basis of this object, a solid is created (by adding thickness) and more solid modelling operations are performed (adding / subtracting material, chamfers, fillets). The socket suspension is generated similarly. The whole solid is presented in Fig. 7b.

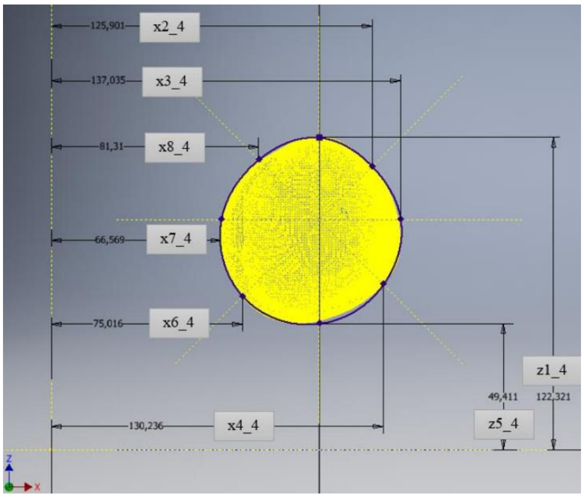


Fig. 6. Closed 8-point spline representing the single section of patient's stump

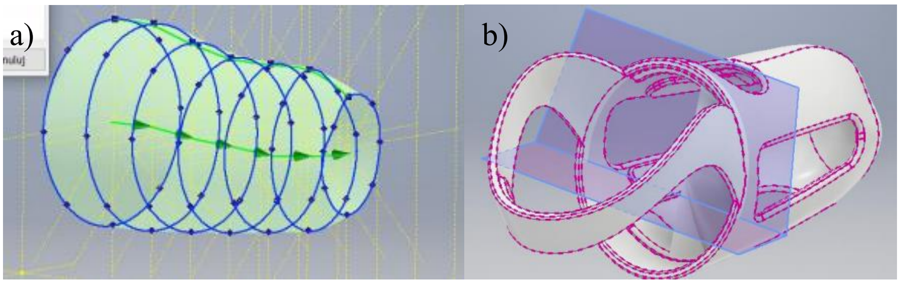


Fig. 7. Multi-section surface (a) and final solid model of socket with suspension (b)

The second part of the prosthesis – the forearm – was made in a very similar way, also by sections, filtering points and feeding them to an Inventor model with iLogic rules. However, the forearm in a given limb is missing (hence the need for prosthesis), so in both cases the data was taken from scan of the second, healthy forearm. The third part – the prosthetic hand or end effector (Fig. 8) – was designed separately, without using 3D scan data, but rather a given set of parameters for configuration (e.g. length, width, thickness etc.).

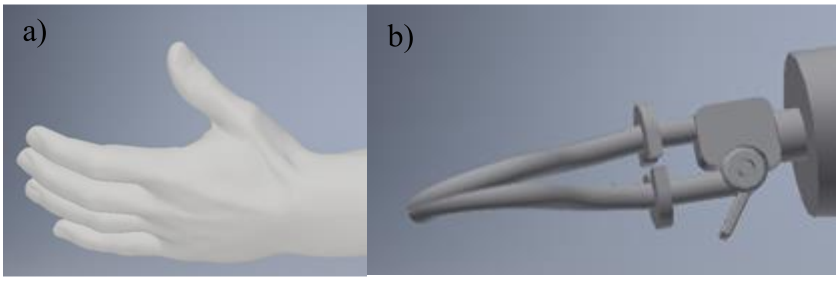


Fig. 8. The designed final parts of the prosthesis, a) cosmetic hand, b) mechanical gripper

The design procedure was performed initially for the patient 1. The second patient's data were used to verify the model correctness and to check if there were any errors. Both obtained models were subjected to digital and physical testing. The digital testing involved fitting with a 3D scanned mesh of patient's limb, to test how it fits and whether the compression and release zones are actually present where intended. The physical testing involved 3D printing of the sockets out of ABS material using an FDM 3D printer and testing basic strength, stiffness and surface quality, as well as manufacturing time. Actual fitting of the sockets involving presence of patients was delayed due to unavailability of the patients at the time of research. Besides the sockets, the prosthetic forearm and hand were also manufactured for the patient 1, forming a complete prosthesis.

2.4. Mechanical prosthesis design

The mechanical prosthesis design was based on the UnLimbited Arm open source project [18]. The original project is already a customizable 3D model, but range of parameters used for its configuration is very limited. Only three parameters are used: length of arm and forearm, as well as circumference of the stump. As results of interviews done with patients and doctors using the prosthesis (realized via direct contact with e-Nable Poland non-profit organization [19]), it was concluded that the prostheses are not precisely adjusted, often being too unbalanced for the users or offering too weak grip. Hence, it was decided to build an intelligent model fed with data measured on a 3D scan (rather than directly on a patient), containing more than a dozen of parameters for customization. The details of the creation of the CAD model itself are presented in [20].

Figure 9 presents a view of spreadsheet with possible parameters – the mechanical hand itself is a separate assembly and has a separate set of parameters, so it can be used as an independent prosthesis of RoboHand type [21]. Figure 10 presents a view of a complete, exemplary prosthesis.

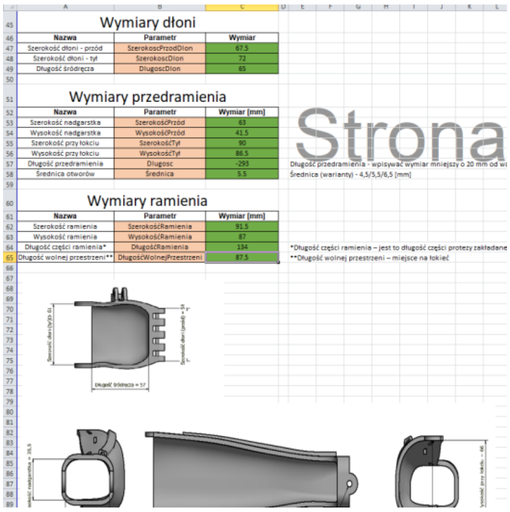


Fig. 9. Parameters of mechanical prosthesis – spreadsheet (fragment)

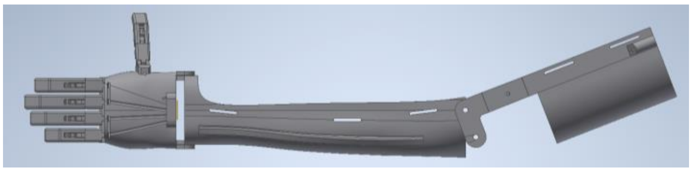


Fig. 10. Exemplary complete prosthesis 3D model

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The model itself was tested for correctness using two sets of parameters. Firstly, a number of upper limbs of healthy adult patients was measured and introduced. Secondly, 3 sets of parameters for child patients were taken from atlas containing anthropometric data [22]. The resulting errors in models (mostly introduced by the data of child patients) were corrected in the final version of the prosthesis.

The physical models of prostheses were manufactured for both patient 1 and 2. In the case of patient 1, the data were taken from available 3D scans. Fitting of the sockets was delayed due to patient unavailability.

In the case of patient 2, two prostheses were made: a traditional UnLimbited Arm, configured via self-measurement (parameters sent by the patient remotely) and a version based on 3D scan. It was possible to verify the prosthesis operation in patient's presence. The prostheses were 3D printed using low-cost FDM process, out of PLA and ABS materials.

3. Results

3.1. Digital testing results

In terms of cosmetic prosthesis, namely prosthetic socket, the verification for patient 2 brought mostly positive results. No direct errors were observed, the basic shape of the socket was generated properly. However, some parameters needed manual adjustment for correctness – chamfers, fillet radiuses, hole sizes etc. The difference was due to both size and age difference between two patients. In the future, these parameters can be automatically selected on the basis of patient age, height and (possibly) weight or other parameters easily obtained or measured during interview.

Another observed problem for patient 2 was related with the 3D scan itself – the mesh was not given a correct coordinate system during the semi-automated MeshLab software processing. This resulted in a necessity of adjustment in some point-selecting functions and formulas governing the basic splines, to compensate the slight shift in coordinate system. In the future, all the scans should be made on the exact same work stand, with the same coordinate system.

The created prostheses were subjected to collision analysis. For both patient 1 and 2, it was found that shape of the socket properly covers the shape of the stump, while the compressing insets collide with it in proper places, according to requirements of CRS/TRAC sockets (Fig. 11).

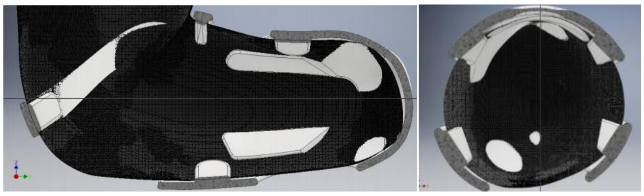

Fig. 11. Verification of geometry of prosthetic socket for patient 1

Regarding the mechanical prosthesis, the verification again was mostly positive. There were some errors in generation of a proper geometry, mostly as a result of wrong measurements in the palm area, performed by patients themselves. An example of such errors are presented in Fig. 12. The model was corrected accordingly and was properly generated for both test patients on the basis of 3D scan measurements. Still, it was noted that two separate versions of the model would need to be created, for adult and child patients, to limit the range of possible errors.

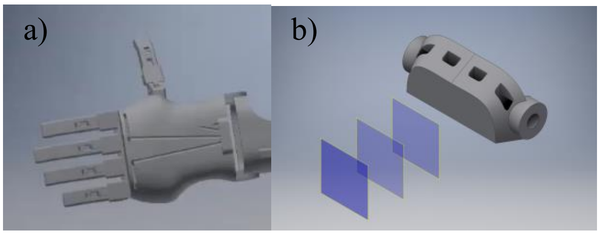

Fig. 12. Errors in automatically generated mechanical hand prosthesis, a) improper palm shape (not proportional dimensions), b) no palm shape (too low dimension value entered)

3.2. 3D printing and testing results

The prosthetic sockets were successfully 3D printed (Fig. 13a), during the time of several hours each (patient 1 – 9,5 hours, patient 2 – 5 hours). The surface was rated as "standard", needing mechanical or chemical treatment for better quality. The accuracy was not evaluated at the time of preparing the paper, however it is planned in the future. For patient 1, prosthetic hands of various materials (PLA, ABS, TPU), as well as forearms were manufactured (Fig. 13b) to obtain complete prosthesis. Time of obtaining a complete prosthesis was approx. 20 hours for patient 1 and approx. 13 hours for patient 2. Satisfying visual and mechanical properties were obtained in the simple operation tests, with maximum strength value at break of hand fingers (bending test) recorded at 500 N. Cost of the complete prosthesis did not exceed 1000 PLN (~250 USD).

The mechanical prostheses were also 3D printed successfully. Complete prostheses were obtained after 10 hours for patient 1 and 6 hours for patient 2, with total cost not exceeding 300 PLN (~75 USD). Visual and mechanical results were acceptable for the PLA material, with fingers sustaining maximally 1000 N and palm sustaining maximally 1200 N of bending force. Figure 14 presents the complete prosthesis for patient 1. For patient 2, the basic, non-modified UnLimbited Arm prosthesis made on the basis of self-measuring of 3 basic dimensions was far too large to be useful. The second version, where measurement was made by the researchers was fitting the patient quite fine.

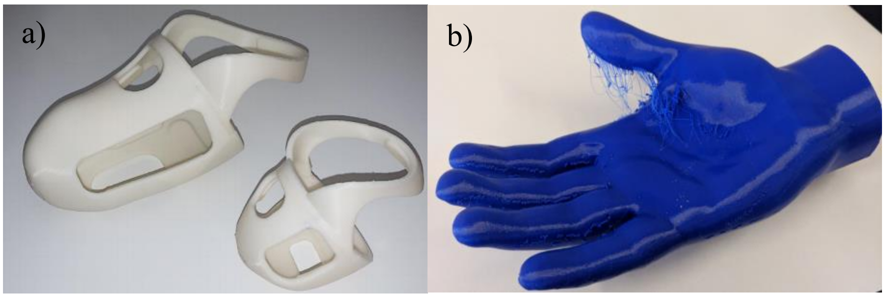

Fig. 13. 3D printed sockets, PLA material (a) and hand, TPU material (b), before post processing treatment

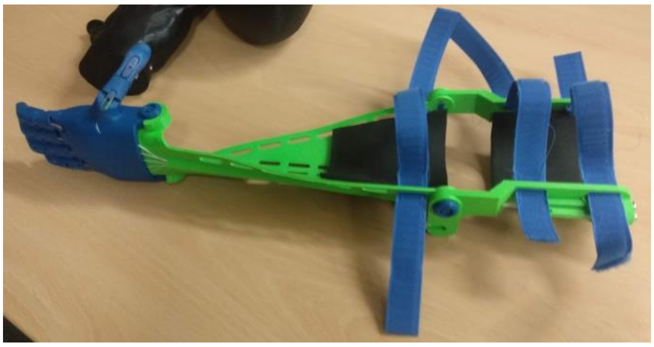

Fig. 14. 3D printed complete mechanical prosthesis for patient 1

4. Conclusions

The obtained results can be described as very promising. The automatic generation of 3D models of prosthetic sockets and complete prostheses, both mechanical and cosmetic, is surely feasible on the basis of 3D scan, both for adults and younger patients, using the presented approach with parametric representation. There are certain problems that need to be overcome, stated in the results section (better consideration of patient's age and general size, common systems of coordinates of 3D scans, problematic self-measuring by the patients). However, the benefit of obtaining a ready-for-print 3D model of such a complex, demanding product as an upper limb prosthesis, in a time of mere minutes – as time between starting a 3D scan and obtaining a print-ready model is between 15 and 45 minutes, the latter only if corrections are needed, which is assumed untrue in future, improved versions – is invaluable.

The low-cost 3D printers make it possible to easily obtain valuable, usable prostheses by the fraction of cost of the commercial, professional solutions, provided that the design itself does not raise costs. That is why it is important to automate the design tasks, to make it available for the wider public. Nowadays, the younger patients often do not receive prostheses, as they cost much and take long to be manufactured and delivered, which is very demanding for the parents (as the prostheses must be exchanged every several months due to child growth). There is also a problem with rapid supplying of adult accident victims with temporary cosmetic or mechanical prostheses, to make them used to such a device, due to the same reasons as with the child patients: time and money. The presented approach can solve these problems, provided it will be universal enough to allow correct generation of prostheses for 95% or more patients. For this, more studies are needed with real patients, which is exactly what authors are planning to do in the near future.

Acknowledgements

The studies were realized with a support from Polish National Center for Research and Development, in the scope of the "LIDER" program (grant agreement no. LIDER/14/0078/L-8/16/NCBR/2017).

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